Tuple Creation

Empty tuple

In [2]:
```
tup1=()
tup1
```

Out[2]:
```
()
```

Tuple of integers

In [3]:
```
tup2=(10,30,60)
tup2
```

Out[3]:
```
(10, 30, 60)
```

Tuple of float

In [4]:
```
tup3=(10.77,30.66,60.89)
tup3
```

Out[4]:
```
(10.77, 30.66, 60.89)
```

Tuple ofstrings

In [6]:
```
tup4=('one','two',"three")
tup4
```

Out[6]:
```
('one', 'two', 'three')
```

Nested tuples

In [7]:
```
tup5=('Asif',25,(50,100),(150,90))
tup5
```

Out[7]:
```
('Asif', 25, (50, 100), (150, 90))
```

Tuple of mixed datatypes

In [9]:
```
tup6=(100, 'Asif',17.765)
tup6
```

Out[9]:
```
(100, 'Asif', 17.765)
```

In [10]:
```
tup7=('Asif', 25,[50,100],[150,90],{'John','David'},(99,22,33))
```

In [11]:
```
len(tup7)
```

Out[11]: 6

Tuple indexing

Retrive first element of the tuple

In [13]:
```
tup2[0]
```

Out[13]: 10

In [14]:
```
tup4[0]
```

Out[14]: 'one'

Nested indexing-Access the first character of the first tuple element

In [15]:
```
tup4[0][0]
```

Out[15]: 'o'

In [16]:
```
tup5[0][0]
```

Out[16]: 'A'

Last item of the tuple

In [17]:
```
tup4[-1]
```

Out[17]: 'three'

In [18]:
```
tup5[-1]
```

Out[18]: (150, 90)

In [20]:
```
tup6[-1]
```

Out[20]: 17.765

Tuple slicing

In [22]:
```
mytuple=('one','two','three','four','five','six','seven','eight')
mytuple
```

Out[22]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')

In [23]:
```
mytuple[0:3]
```

Out[23]: ('one', 'two', 'three')

In [24]:
```
mytuple[2:5]
```

Out[24]: ('three', 'four', 'five')

In [25]:
```
mytuple[:3]
```

Out[25]: ('one', 'two', 'three')

In [26]:
```
mytuple[:2]
```

Out[26]: ('one', 'two')

In [27]:
```
mytuple[-3:]
```

Out[27]: ('six', 'seven', 'eight')

In [29]:
```
mytuple[-2:]
```

Out[29]: ('seven', 'eight')

In [30]:
```
mytuple[-1:]
```

Out[30]: ('eight',)

In [31]:
```
mytuple[:]
```

Out[31]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')

Remove and change items

In [32]:
```
mytuple
```

Out[32]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')

tuples are immutable which means we cannot delete tuple items

In [35]:
```
del mytuple[0]
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[35], line 1
----> 1 del mytuple[0]

TypeError: 'tuple' object doesn't support item deletion
```

tuples are immutable which we cannot change tuple items

In [36]:
```
mytuple[0]=1
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[36], line 1
----> 1 mytuple[0]=1

TypeError: 'tuple' object does not support item assignment
```

Deleting entire tuple object is possible

In [37]:
```
del mytuple
```

Loop through a tuple

In [39]:
```
mytuple=('one','two','three','four','five','six','seven','eight')
```

In [40]:
```
mytuple
```

Out[40]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

In [43]:
```
for i in mytuple:
    print(i)
```

```
one
two
three
four
five
six
seven
eight
```

In [44]:
```
for i in enumerate(mytuple):
    print(i)
```

```
(0, 'one')
(1, 'two')
(2, 'three')
(3, 'four')
(4, 'five')
(5, 'six')
(6, 'seven')
(7, 'eight')
```

Tuple membership

In [45]:
```
mytuple
```

Out[45]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

checking if one exists in the list

In [46]:
```
'one' in mytuple
```

Out[46]:
```
True
```

In [47]:
```
'ten' in mytuple
```

Out[47]:
```
False
```

checking if three exists in the list

In [50]:
```
if 'three' in mytuple:
    print('three is present in the tuple')
else:
    print('three is not present in the tuple')
```

```
three is present in the tuple
```

checking if eleven exists in the list

In [51]:
```
if 'eleven' in mytuple:
    print('eleven is present in the tuple')
else:
    print('eleven is not present in the tuple')
```

```
eleven is not present in the tuple
```

Index position

In [52]:
```
mytuple
```

Out[52]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

index of first element is equal to 'one'

In [53]:
```
mytuple.index('one')
```

Out[53]:
```
0
```

In [54]:
```
mytuple.index('five')
```

Out[54]:
```
4
```

In [59]:
```
mytuple1=('one','two','three','four','one','one','two','three')
mytuple1
```

Out[59]:
```
('one', 'two', 'three', 'four', 'one', 'one', 'two', 'three')
```

In [60]:
```
mytuple1.index('one')
```

Out[60]: 0

In [61]:
```
mytuple.index('four')
```

Out[61]: 3

Sorting

In [63]:
```
mytuple2=(43,67,99,12,6,90,67)
mytuple2
```

Out[63]: (43, 67, 99, 12, 6, 90, 67)

Returns a new sorted list and doesn't change original tuple

In [64]:
```
sorted(mytuple2)
```

Out[64]: [6, 12, 43, 67, 67, 90, 99]

sort in descending order

In [65]:
```
sorted(mytuple2, reverse=True)
```

Out[65]: [99, 90, 67, 67, 43, 12, 6]

In [ ]:

In [ ]:

In [ ]: